Grounded Multimodal Retrieval & Evidence-Based Report Generation

Multimodal Retrieval Systems Engineering for Disaster Intelligence Applications

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GitHub Repository: https://github.com/Johnson-Opadere/grounded-multimodal-retrieval

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Disaster Intelligence Requires More Than Generation

Challenges

Information Overload

Fragmented image, text and audio evidence

Limited evidence provenance

Hallucination-prone generation

Project Goal

Query Image

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Multimodal Retrieval

↓

Ground Evidence

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LLM Report Generation

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Reliable Disaster Assessment

Project 2B combines multimodal retrieval, evidence grounding, and LLM generation to produce traceable disaster assessments.

Project Objectives

✓ Operationalize multimodal embeddings

✓ Build FAISS-based retrieval infrastructure

✓ Quantify retrieval performance

✓ Develop metadata-aware retrieval policies

✓ Enable grounded report generation

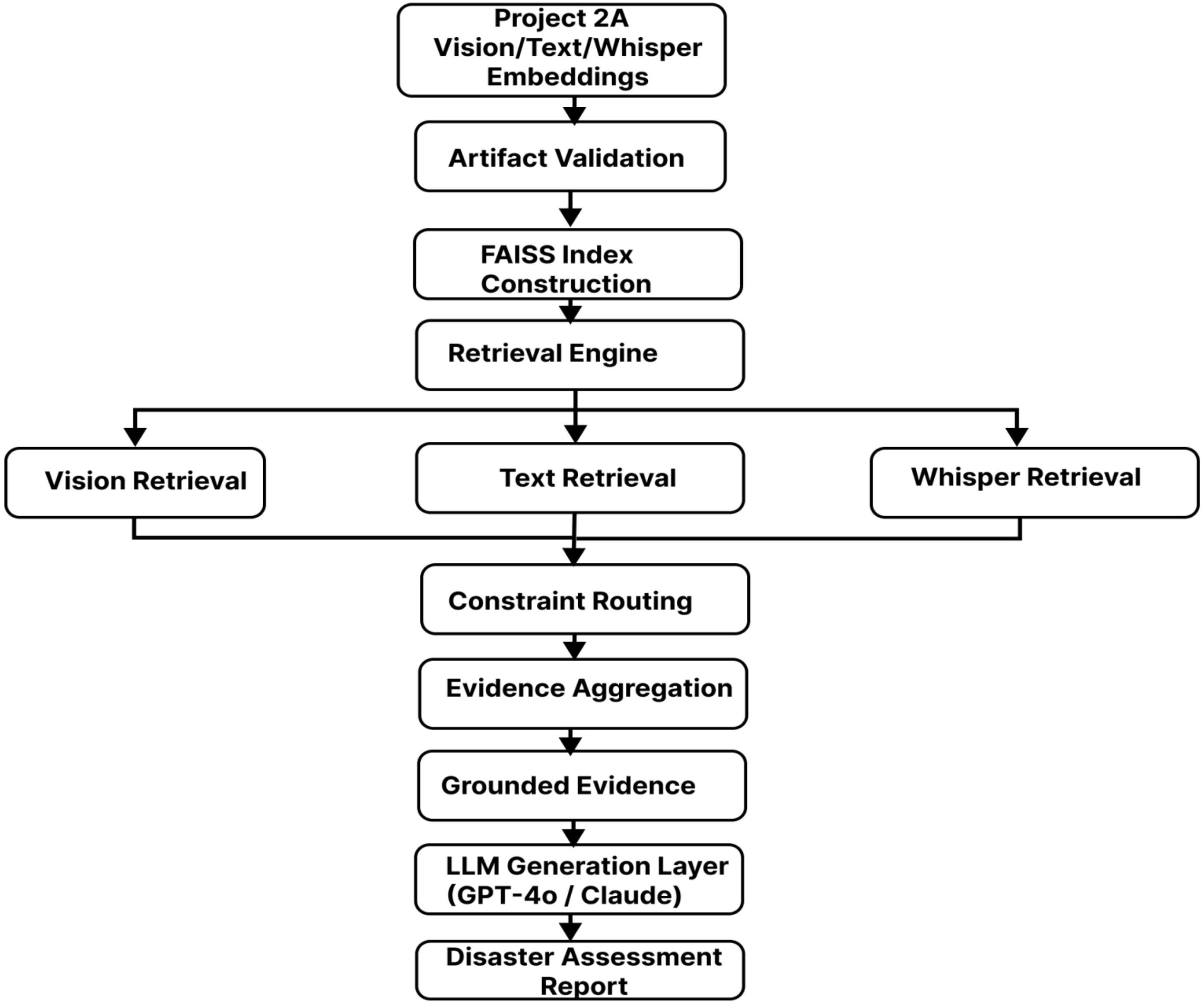
✓ Generate evidence-based assessments

Project 2B transforms learned multimodal embeddings into a deployable retrieval and grounding pipeline.

System Architecture

Key Takeaway

- Project 2A learned multimodal embeddings
- Project 2B operationalizes those embeddings through retrieval, grounding, and report generation

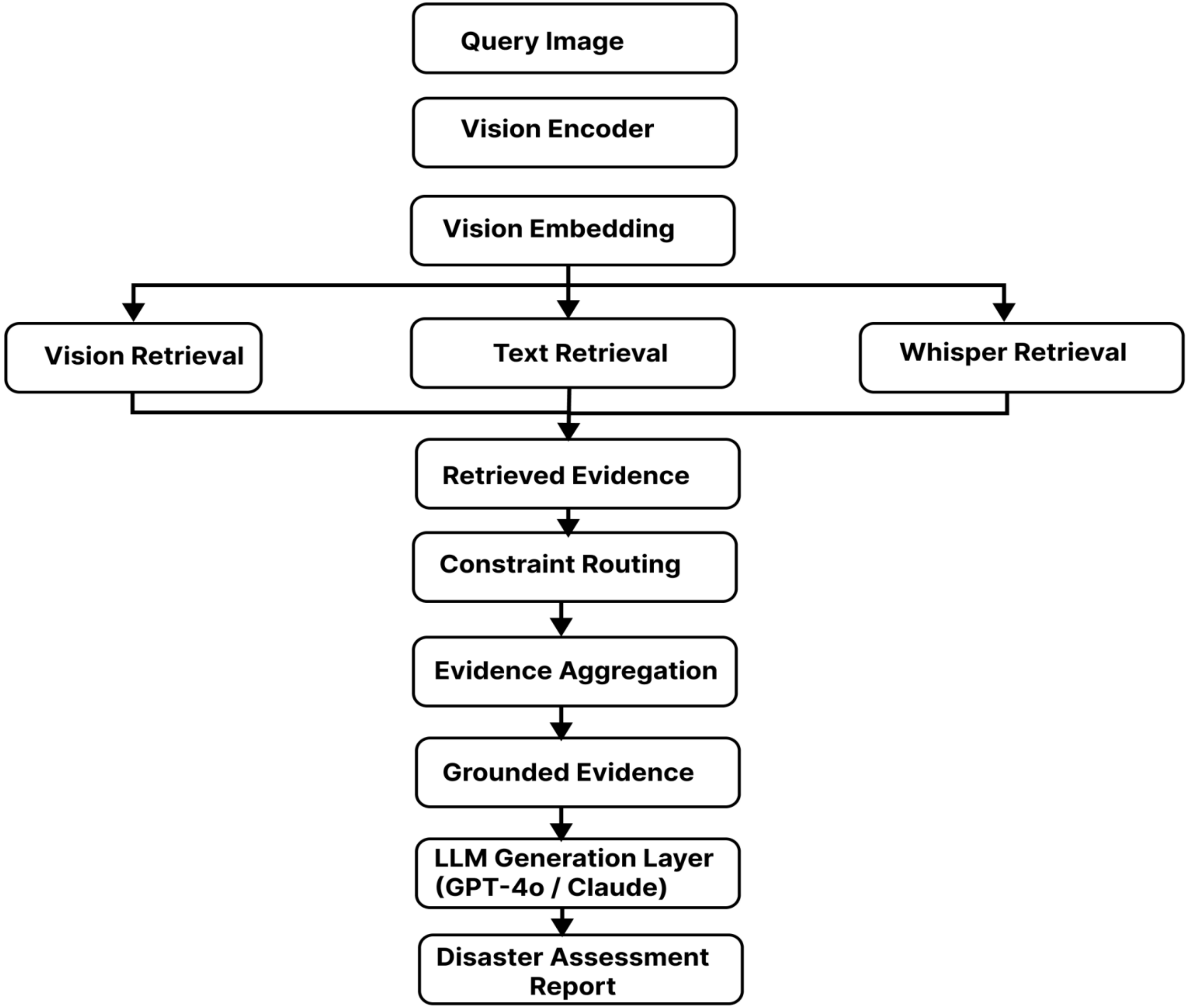


Project 2A learned multimodal embeddings. Project 2B operationalizes them through retrieval and grounding.

Retrieval Pipeline

Key Takeaway

- Multimodal retrieval transforms a query image into grounded evidence for reliable report generation



Multimodal retrieval transforms a query image into grounded evidence for report generation.

Metadata-Aware Retrieval Policies

Key Takeaway

Multimodal retrieval transforms a query image into grounded evidence for reliable report generation

Mode	Purpose
none	Baseline retrieval
same_bucket	Operational retrieval
disaster_family	Balanced diversity
same_event	Maximum consistency

Retrieval constraints improve retrieval coherence and operational stability.

Learned Visual Embedding Space

Result: same_bucket_ratio = 0.9790

The learned visual embedding manifold captures meaningful disaster semantics across events.

Key Observations

Cross-event similarity

Semantic clustering

Visual consistency

Constraint-Based Retrieval

Unrestricted vs Constrained Retrieval

MODE: none (Unrestricted Retrieval)

[1] (volcanic_damage) midwest-flooding score=0.0366 countless roads and bridges are washed out and/or damaged in nebraska, cutting off whole communities. -- [2] (volcanic_damage) guatemala-volcano score=0.0246 flows of mud and ash have covered roads and crops and destroyed bridges. -- [3] (wildfire) santa-rosa-wildfire score=0.0137 stags leap district: hagafen (minor damage) and signorello estate (winery and hospitality center completely burned down) and some buildings at stags' leap winery. -- [4] (generic_damage) hurricane-michael score=0.0113 you cannot tell where he stopped harvest and where the rest of it was being harvested by the storm." more than poultry houses were also destroyed, officials say. -- [5] (generic_damage) hurricane-matthew score=0.0100 , homes damaged or destroyed breached dams ,, cubic yards of storm debris , household power outages copyright wmbf. --

MODE: same_bucket (Operational Stabilization)

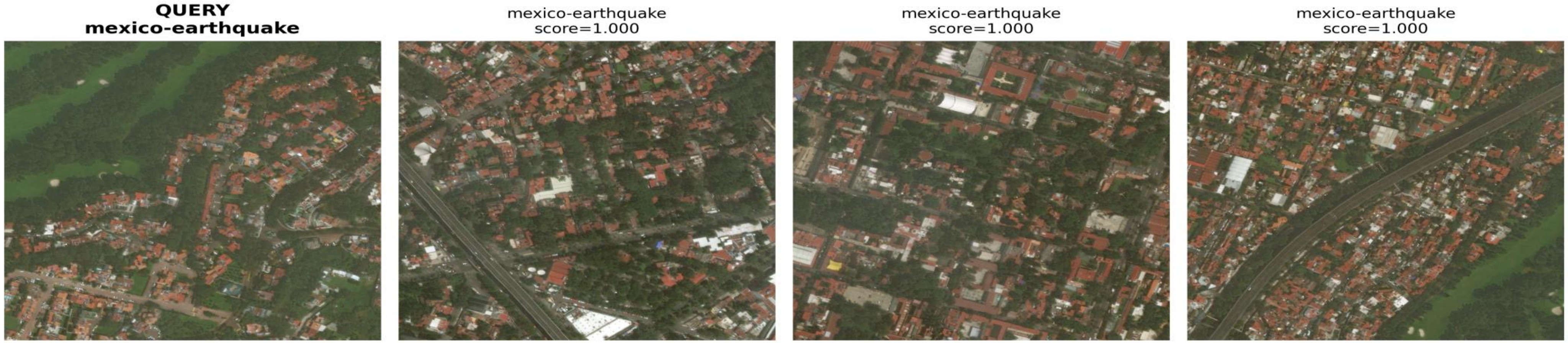
[1] (flooding) hurricane-florence score=-0.0269 significant flooding destroyed homes and businesses in low-lying parts of lumberton. -- [2] (flooding) hurricane-florence score=-0.0309 in the town of nichols approximately homes recently rebuilt after flooding from hurricane matthew were damaged again. -- [3] (flooding) hurricane-florence score=-0.0339 flooding damaged approximately homes throughout the county. -- [4] (flooding) hurricane-florence score=-0.0366 although florence will be remembered primarily for its record-breaking flooding, wind gusts over mph caused significant damage to buildings, trees, and electrical service across the cape fear area, and a storm surge of over four feet eroded beaches and damaged property between cape fear and cape lookout. --

Key Takeaway

Metadata-aware constraints improve retrieval coherence and operational stability.

Operational retrieval policies improve evidence quality without modifying the underlying embedding space.

Grounded Evidence Retrieval



Retrieved Text Evidence

[1]
teams formed to dig through rubble — which at the time included hospitals, apartments and hotels.

[2]
thousands of homes have collapsed, along with hospitals, hotels and a shopping centre.

Grounded Operational Summary

Retrieved evidence suggests severe flooding-related damage across multiple disaster events.

Observed impacts include:
- damaged roads and bridges
- community isolation
- transportation disruption

Retrieved visual neighbors demonstrate cross-event disaster similarity.

Components

Image + Retrieved Text + Operational Summary

Key Takeaway

Evidence aggregation enables grounded reasoning

Multimodal evidence is consolidated into a grounded context for downstream report generation.

Retrieval Evaluation

Vision → Vision

same_bucket: 0.979

same_event: 0.773

Vision → Text

same_bucket: 0.015

same_event: 0.063

Cross-Modal Retrieval

Recall@5: 0.0648

MRR: 0.0153

XE: 0.0173

Key Takeaway

Visual retrieval is highly coherent, while cross-modal retrieval remains challenging and motivates future work.

Strong visual retrieval validates the learned embedding space, while cross-modal alignment remains an open problem.

Grounded Evidence Enables Reliable Report Generation

Retrieval Pipeline

Retrieved Evidence

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Grounded Prompt

↓

GPT-4o /Claude

↓

Disaster Report

Example Generated Report

Likely disaster type
Hurricane-driven flooding (retrieval-derived)

Observed Damage

- Infrastructure damage
- Housing destruction

Operational Implications

- Access disruption
- Community isolation

Retrieved evidence provides provenance for generated conclusions and reduces unsupported reasoning.

Key Takeaway

Grounding enables traceable and evidence-based disaster assessment reports

Key Findings & Future Directions

Key Findings

- ✓ Vision embeddings learned meaningful disaster semantics
- ✓ Cross-modal retrieval remains challenging
- ✓ Metadata-aware routing improves stability
- ✓ Grounding improves report reliability

Future Work

- ✓ Cross-encoder reranking
- ✓ Larger evidence corpora
- ✓ Geospatial retrieval
- ✓ Real-time disaster monitoring

Project 2B demonstrates multimodal retrieval, grounded AI, and evidence-based report generation for disaster-response applications